

Phase 1 - Week 3 - Lesson Day 2

Seaborn & Plotly

Today's Objectives

Matplotlib gives you full control but requires more code for polished results. Today introduces two libraries built to speed that up: **Seaborn** (built on top of Matplotlib, designed for statistical charts with far less code) and **Plotly** (designed for interactive charts you can hover, zoom, and explore - closer to what Power BI/Tableau dashboards feel like).

- Build statistical charts quickly with Seaborn
- Build a heatmap to visualize correlations between columns
- Build a pairplot to see relationships across many columns at once
- Build an interactive chart with Plotly

1. Seaborn Basics

Definition: Seaborn is a plotting library built on top of Matplotlib, designed specifically for statistical visualization. It works especially well directly with Pandas DataFrames, and produces more polished-looking charts with less code than raw Matplotlib.

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.histplot(data=df, x="age", bins=15) # histogram, Seaborn style
plt.show()

sns.boxplot(data=df, x="city", y="age") # box plot - shows spread AND outliers per category
plt.show()
```

Box plots (new concept): a box plot shows the median (middle line), the middle 50% of values (the box), and any outliers (dots beyond the "whiskers"). It's a compact way to compare the spread of a numeric column across several categories at once - something a histogram alone can't easily do.

2. Heatmaps

Definition: A heatmap uses color intensity to represent values in a grid - most commonly used in data science to visualize a **correlation matrix**, showing how strongly every numeric column relates to every other numeric column at a glance.

```
correlation_matrix = df.corr(numeric_only=True) # calculates correlation between all numeric columns

sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm")
```

```
# annot=True prints the actual number on each cell
# cmap sets the color scheme
plt.show()
```

Correlation values range from -1 to 1. Values near **1** mean two columns move strongly together (as one goes up, so does the other). Values near **-1** mean they move in opposite directions. Values near **0** mean there's little to no linear relationship.

3. Pairplots

Definition: A pairplot draws a scatter plot for *every pair* of numeric columns in your dataset at once, arranged in a grid, with histograms along the diagonal. It's the fastest way to spot relationships and patterns across many columns without plotting each pair manually.

```
sns.pairplot(df)
plt.show()

# Color the points by a categorical column to spot group patterns
sns.pairplot(df, hue="city")
plt.show()
```

hue colors each point by a category, which often reveals patterns a plain pairplot would hide - for example, two groups that each show a relationship, but in opposite directions.

4. Plotly (Interactive Charts)

Definition: Plotly builds charts you can interact with directly in the notebook - hovering shows exact values, and you can zoom, pan, and click to filter. This is much closer to the experience of exploring a Power BI or Tableau dashboard than a static Matplotlib image.

```
import plotly.express as px

fig = px.scatter(df, x="age", y="income", color="city",
                 title="Age vs Income by City")
fig.show()
```

plotly.express (imported as **px**) is the quick, high-level interface to Plotly - almost every common chart type (scatter, bar, line, histogram) is available as a single function call, similar in spirit to Seaborn but with built-in interactivity.

Today's Exercise

Open **day-2-seaborn-plotly.ipynb**. As with Day 1, every exercise is fully solved using generated sample data so the notebook runs immediately. Run each cell and review the comments.

Exercise Questions (4 total - all solved in the notebook)

1. Seaborn box plot

Create a box plot comparing a numeric column's spread across 3-4 categories.

2. Correlation heatmap

Build a sample DataFrame with at least 4 numeric columns, calculate the correlation matrix, and plot it as a heatmap with values shown.

3. Pairplot with hue

Build a sample DataFrame with 3 numeric columns and 1 categorical column, and create a pairplot colored by the categorical column.

4. Interactive Plotly scatter

Create an interactive scatter plot with Plotly, coloring points by a categorical column.

Once you and Halimo have reviewed all four solved exercises, move on to Lesson Day 3 (Descriptive Statistics).